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1 Justification

The capability of telemetry to transmit real-time data efficiently in critical systems is significantly challenged within the evolving landscape of Industry 4.0 [1]. Telemetry, which serves as the backbone for transmitting crucial information in many industrial and vehicular applications, faces severe limitations when applied to dynamic environments. These challenges are particularly evident in mobility-based systems such as Vehicle-to-Vehicle (V2V) communication [2], where devices are often traveling at high speeds.

In such scenarios, the need for reliable, high-speed data transmission becomes important. V2V systems, typically may necessitate data rates ranging from 2 to 54 Mbps at a distance of 15 to 1000m, with a maximal vehicular speed of 200 km/h [4].

Autonomous vehicles depend heavily on uninterrupted, real-time data exchange to make split-second decisions, and any delays or inaccuracies in data transmission could result in dangerous situations [3] and could result in the complete loss of equipment or even pose a danger to personnel

Doppler effect and latency will cause distortions in the communication system and demand retransmission or reading of corrupted data [5]. These elements taken together provide a significant obstacle to reach: quick fading channels, fast mobility, and high-speed data demand.

References

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